

AI-Powered Logistics Operations Automation

**Rate Confirmation Processing • UiPath • Document Understanding •
AI Workflow Automation**

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Business Context

Carrier operations teams process large volumes of broker rate confirmations daily. These documents contain critical shipment, rate, pickup, delivery, and reporting information needed for operational tracking and profitability analysis.

Manual handling of rate confirmations creates repetitive workflow bottlenecks across document review, shipment data extraction, mileage lookup, spreadsheet reporting, and RPM calculations.

Operational Challenges

- Manual review of broker rate confirmation PDFs
- Repetitive extraction of shipment details from inconsistent document formats
- Separate mileage lookup process using external tools
- Manual spreadsheet entry for reporting and RPM calculations
- Limited visibility into shipment profitability and operational performance
- Difficult to scale reporting workflows as document volume increases

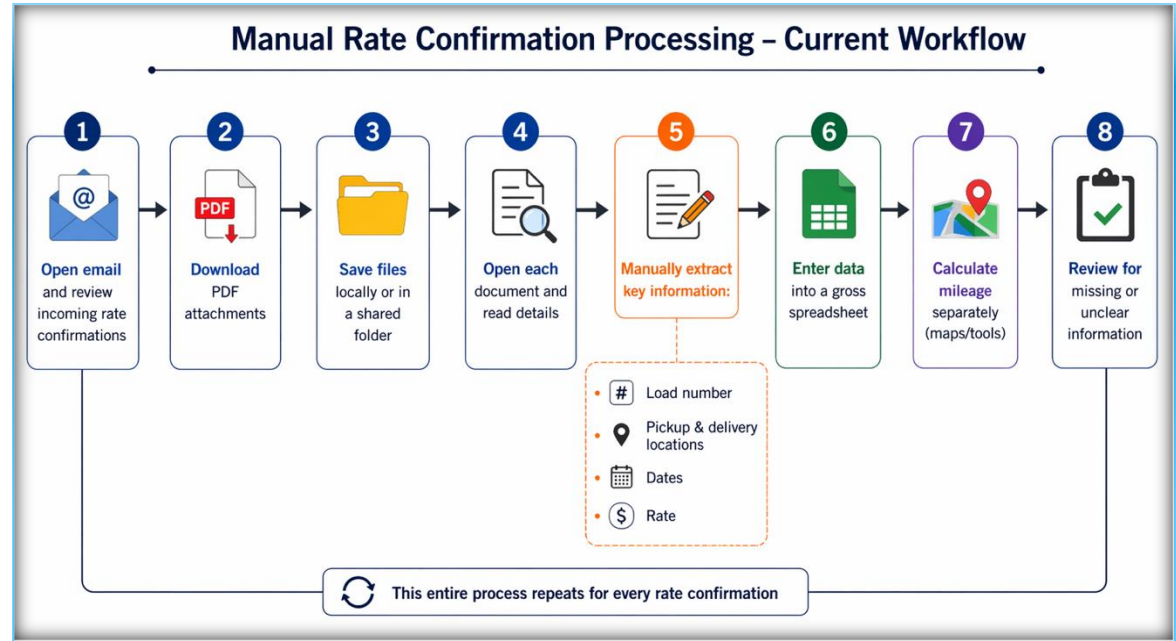
Business Impact

- ~5 hours/week spent on repetitive document processing and reporting tasks
- Increased risk of manual data entry errors
- Delayed RPM and profitability reporting
- Reduced operational visibility across shipment workflows
- Limited scalability of manual carrier operations processes

Manual Rate Confirmation Workflow

The original workflow required manual handling of each rate confirmation, including downloading documents, reviewing shipment details, calculating mileage separately, and entering data into a reporting spreadsheet.

This created a repetitive, time-consuming process that slowed reporting and made operational performance analysis harder to scale.



Automation Solution

An AI-powered logistics operations workflow was developed in UiPath to streamline rate confirmation processing through automated document extraction, mileage enrichment, validation workflows, and structured operational reporting.

The solution transforms unstructured shipment documents into structured operational data, improving reporting efficiency, workflow scalability, and operational visibility across carrier operations.

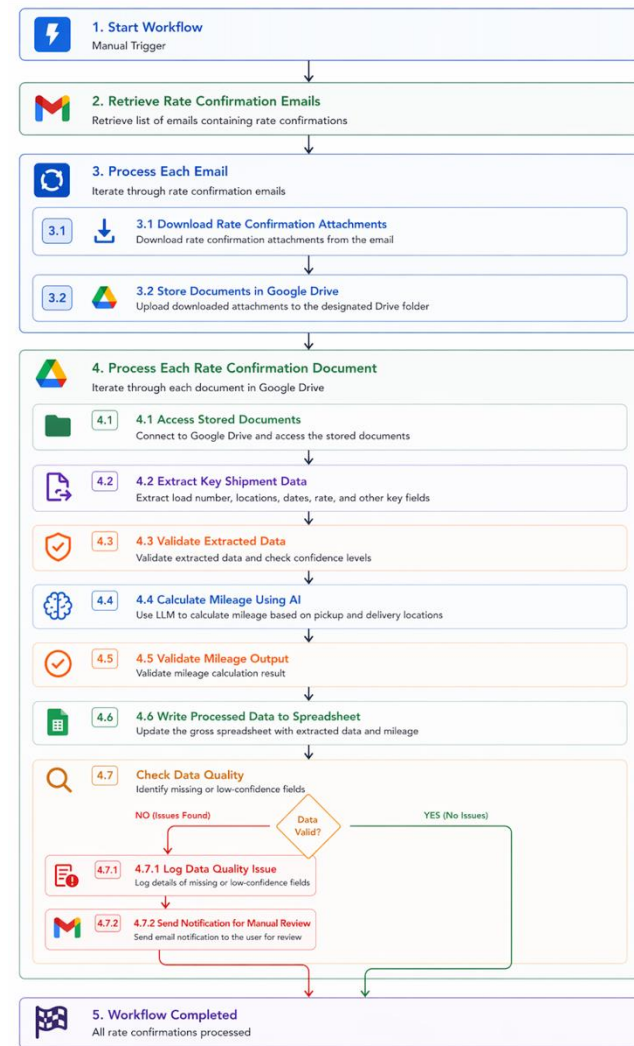
Core Capabilities:

- Automatically retrieves rate confirmations from operational email workflows
- Stores and organizes documents in Google Drive
- Extracts shipment data from unstructured PDF documents using Document Understanding
- Calculates mileage using AI-based enrichment
- Writes structured reporting outputs into Google Sheets
- Flags incomplete or missing data for manual review
- Sends automated validation alerts when issues are detected
- Improves reporting visibility and reduces repetitive manual processing

Automation Workflow Architecture

This workflow automates the end-to-end processing of logistics rate confirmations, from email intake and document extraction to mileage calculation, validation, and structured reporting.

The automation connects multiple operational steps into a single workflow, reducing repetitive manual processing and improving reporting efficiency.



Automation Architecture Components

The automation was implemented in UiPath by combining multiple components, including email integration, file handling, document understanding, AI-based enrichment, and validation logic.

Key Components:

- Email integration to retrieve rate confirmations
- Google Drive integration for document storage and organization
- Document Understanding pipeline for data extraction
- AI integration for mileage calculation
- Validation logic for data quality control
- Google Sheets integration for structured reporting output

Document Understanding & Custom Extractor

A custom generative extractor was configured to extract shipment data from broker rate confirmations, which often vary in structure and formatting.

The workflow was designed to process unstructured logistics documents and convert them into structured operational data for reporting and analysis.

Field name	Instruction
Name of the field to extract from the document	Instruction about what information to extract about the field
Load number	Extract the load number from the rate confirmation. This is the broker reference identifying the load. Return only the load number as written in
Pickup location	Extract the pickup location for the load. Return city and state only, and zip code if available.
Delivery location	Extract the delivery location for the load. Return city and state only, and zip code if available.
Pickup date	Extract the scheduled pickup date for the load. Return date in mm/dd/yy format.
Total rate	Extract the total agreed carrier pay or total rate for this load. Return numbers only in 0.00 format without currency symbol or commas.

Extract Key Shipment Data

DU models not currently defined as part of Solution definitions. Please make sure they are created manually and exist when deploying the solution to other tenants.

Input*

{x} CurrentItem x

Document Understanding project *

Generative Predefined x v

Extractor *

Long Document Simple Layout Extractor x v

Document Type Details*

7 items in collection

AI-Powered Mileage Calculation

LLM-based mileage estimation was implemented as a lightweight AI enrichment approach during initial workflow prototyping.

Key Points:

- Uses pickup and delivery locations extracted from documents
- Sends structured prompt to LLM
- Returns estimated mileage for each trip
- Eliminates manual lookup in tools like Google Maps
- Ensures consistent and fast calculation

The workflow automatically calculates mileage without manual lookup, helping simplify reporting and reduce repetitive operational tasks.

 **Calculate Mileage Using AI** 

 Indexes used in these activities are not currently defined as part of Solution definitions. Please make sure they are created manually and exist when deploying the solution to other tenants.

UiPath GenAI Activities connection

 UiPath GenAI Activities 

Model name

anthropic.claude-haiku-4-5-20251001-v1:0 (Claude)   

Prompt *

You are helping a logistic manager to calculate the miles for each trip.  
Look at the pickup `{x} documentData.Data.PickupLocation` and
delivery `{x} documentData.Data.DeliveryLocation` locations to
calculate the total mileage. The answer must be numbers only.

Data Mapping & Debugging

Extracted shipment data is mapped into a structured output format, with logging used to validate extracted values and support debugging during workflow development.

The workflow helps improve reporting consistency and makes troubleshooting easier when processing different document formats.

Key Points:

- Extracted fields are mapped into structured output columns in Google Sheets
- Log messages are used to validate extracted values before writing output
- Helps identify missing or incorrect data during workflow testing
- Simplifies debugging and improves workflow reliability

 Validate Extracted Data

Message *

Load #: {x} documentData.Data.LoadNumber (confidence: {x} documentData.Data.LoadNumber.Confidence)

Broker name: {x} documentData.Data.BrokerName (confidence: {x} documentData.Data.BrokerName.Confidence)

Log Level

Select a level or leave empty for info

 Write Processed Data to Spreadsheet

Google Sheets connection

@gmail.com_2 X

Spreadsheet*

Maryna Khlynina gross

Use spreadsheet template

Range *

'total gross' (Sheet)

How to write

Append

Has headers

☒ True ☐ False

What to write

Individual Fields

Load #

{x} documentData.Data.LoadNumber

Broker name

{x} documentData.Data.BrokerName

Shipping date

{x} documentData.Data.PickupDate

Validation Logic

Validation logic is used to identify incomplete or low-confidence data before it is written into the final reporting output.

The workflow automatically flags potential issues and sends notifications for manual review when needed.

Key Points:

- Identifies missing or incomplete extracted fields
- Flags low-confidence or suspicious data based on validation rules
- Sends email notifications for manual review
- Prevents incorrect or incomplete data from entering the reporting workflow
- Uses logging and validation checks to improve workflow reliability

This screenshot shows a form titled "Send Notification for Manual Review". It includes a "Gmail connection" field with a dropdown menu showing an email address. Below this is a "Save as draft" section with radio buttons for "True" and "False", where "False" is selected. The "To*" field contains an email address. The "Subject" field is labeled "Rate confirmation requires manual check". The "Body" field contains a message: "Rate confirmation requires manual check! Low confidence on rate confirmation: {x} CurrentItem {x} CurrentItem.Url". The form has a clean, modern design with a light gray background and blue accents.

This screenshot shows a workflow configuration titled "Check Data Quality". The "Condition *" field contains a logical expression: "documentData.Data.LoadNumber.Confidence < 0 OR documentD...". Below this is a "Then" section with a "Log Data Quality Issue" action. The "Message *" field for this action contains the text: "Rate confirmation requires manual check: low confidence {x} CurrentItem". The "Log Level" is set to "Warn". The workflow is visualized with a flowchart showing the condition leading to the logging action.

Workflow Monitoring & Alerts

Workflow logs and automated notifications provide visibility into processing results and help identify records that require manual review.

The monitoring flow highlights incomplete or low-confidence extractions, allowing issues to be reviewed before final reporting output is generated.

Key Points:

- Workflow logs display extracted shipment data and validation results
- Low-confidence records are automatically flagged for review
- Email notifications are triggered when potential issues are detected
- Monitoring output improves visibility into workflow processing
- Helps identify extraction issues before data is written into reports

 11:10:52 PM Load #: (confidence: -1)
Broker name: ADM LOGISTICS, INC...
Pick up date: 01/27/20 (confidence:...
Pick up: DES PLAINES IL 60018 (con...
Delivery date: 01/28/20 (confidenc...
Delivery: NASHVILLE TN 37210 (con...
Rate: 1650.00 (confidence: 0)

 11:10:53 PM mileage: 520

 11:10:53 PM Rate confirmation requires manua...

 11:11:10 PM Load #: 0505637 (confidence: 0)
Broker name: Transplace Texas, LP ...
Pick up date: 01/28/20 (confidence:...
Pick up: ALSIP, IL 60803 (confidenc...
Delivery date: 01/30/20 (confidenc...
Delivery: LOXLEY, AL 36551 (confid...
Rate: 1800.00 (confidence: 0)

 11:11:11 PM mileage: 1047

Rate confirmation requires manual check [Inbox x](#)

  1@gmail.com>
to me ▾

11:10 PM (19 minutes ago) ☆ ☺

Rate confirmation requires manual check! Low confidence on rate confirmation: UiPath.GSuite.Drive.Models.GDriveRemoteItemhttps://drive.google.com/uc?id=1a160HPbcABBHqV_ynExlgQdxAbQq1_E3&export=download

One attachment • Scanned by Gmail ⓘ ✓ Added to Drive 📄 Compress PDF



Structured Reporting Output

A	B	C	D	E	F	G	H	I
Load #	Broker name	Shipping date	Pickup location	Delivery date	Delivery location	Total rate	Miles	RPM
306837	PLS LOGISTICS SERVICES	02/08/23	Botkins, OH 45306	02/10/23	orlando fl 32824	\$3,600.00	1087	
451400H	International Transport Systems, Inc.	03/14/23	LA VERGNE, TN 37086	03/15/23	Pine Bluff, AR 71602	\$1,150.00	289	
CPL808978633	Custom Pro Logistics	01/28/20	Horse Cave, KY 42749	01/29/20	OCONOMOWOC, WI 53066	\$1,300.00	628	
S-0419-1725	HGR Industrial Surplus	01/15/20	Montgomery, IL 60538	01/16/20	Euclid, OH 44117	\$1,300.00	651	
3440KJ	Beard Transport	03/02/23	Blackshear, GA, 31516	03/04/23	Canterbury, NH, 03224	\$3,500.00	1247	
	MAISON LUXE	03/13/23	Upper Sandusky, OH 43351	03/14/23	Ashland City, TN 37015	\$1,500.00	534	
	ADM LOGISTICS, INC.	01/27/20	DES PLAINES IL 60018	01/28/20	NASHVILLE TN 37210	\$1,650.00	512	
505684E	Transplace Texas, LP	01/28/20	ALSIP, IL 60803	01/30/20	LOXLEY, AL 36551	\$1,800.00	1047	
3690729	MERCER TRANSPORTATION CO.	03/13/23	Upper Sandusky, OH 43351	03/14/23	ASHLAND CITY, TN 37015	\$1,623.73	592	
C-213547	Coyote Logistics, LLC	11/12/20	Sterling Heights, MI 48314	11/12/20 - 11/13/	Bellevue, OH 44811	\$1,500.00	286	

Processed shipment data is organized into a structured spreadsheet format, making it easier to review shipment information, identify missing values, and support reporting workflows.

With mileage already calculated, RPM can be generated automatically across all records, reducing manual calculations and improving reporting efficiency.

Key Outcomes

The workflow automates the end-to-end processing of rate confirmations, transforming unstructured shipment documents into structured operational data.

Key Outcomes:

- **~70–80%** reduction in repetitive manual processing
- Automated processing of shipment confirmations
- **AI-based mileage calculation** using extracted location data
- Reduced repetitive data entry and manual lookup
- Faster and more structured **RPM reporting**
- Automatic flagging of incomplete or low-confidence data

Business Impact

Operational Improvements:

- Reduced repetitive manual work
- Improved reporting efficiency
- Increased data consistency
- Supports scalable document processing

Automation Capabilities:

- AI-powered document extraction
- Automated mileage enrichment
- Validation and monitoring workflow
- Structured spreadsheet reporting output

Production Readiness & Future Enhancements

How AI improved it:

- Automated extraction of shipment data from unstructured documents
- AI-based mileage calculation reduces manual lookup work
- Supports processing larger volumes of shipment documents

Future Enhancements:

- Replace LLM mileage estimation with Google Maps API for higher accuracy
- Use a custom-trained extraction model to improve extraction accuracy across different broker formats